

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
IEM 4127
Industrial Management

TIME: 3 hours

FULL MARKS: 210

- N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Necessary chart: Standard Normal Distribution chart will be provided.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Management? Draw a network of management functions and explain its functions. (13)
b) Assume that, you want to promote as "Manager" at 'R-Pac International Limited'. To achieve this role, what managerial skills you need to possess? (12)
c) Explain how management and administration are different from each other? (10)
2. a) Define the contingency theory school. "A successful organization should react as an organic and open system." – Do you agree with this statement? Give logical explanation. (13)
b) Discuss functional, line and line-staff organization briefly. (12)
c) What is delegation? Differentiate between flat structure and tall structure. (10)
3. a) Discuss Herzberg's theory of motivation. How does it differ from Maslow's theory? What are its limitation? (12)
b) Define upper and lower breakeven point. Why breakeven analysis is called shoot run analysis? (10)
c) Shajal is the top businessman in the field of food industry. Pranto is the emerging competitor of Shajal. Both Shajal's and Pranto's company workers are rated in terms of wages 30 paisa per minute. The standard time for the workers is 4.0 hours. Shajal's company workers takes 2.5 hours to complete the job whereas Pranto's company workers take 2.0 hours to complete the job. Calculate the earnings of each company's workers according to Halsey plan. (13)
4. a) What is committee? Mention the condition for effective functioning of committee. (09)
b) Handheld fiber optic meters are useful measuring temperature. The fixed costs associated with manufacturing are \$800,000 per year. If revenue is \$2950 and its variable costs is \$2075, then
i) How many units must be sold for breakeven in each year?
ii) What will be the profit for sales of 3000 units per year? (13)
c) A depreciable construction truck has a first cost of \$20,000 with a \$4000 salvage value after 5 years. Find the (i) Depreciation and (ii) Book value after 3 years using SL and DDB depreciation. (13)

SECTION B

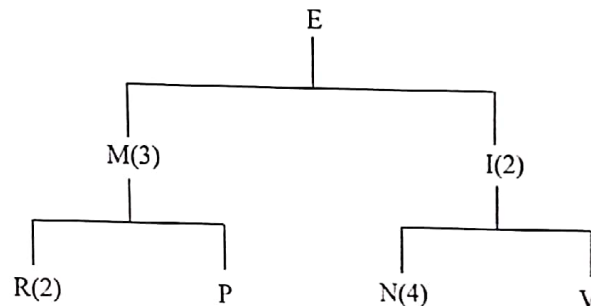
(Answer **ANY THREE** questions from this section in Script B)

5. a) Define forecasting. What is the significance of exponential smoothing technique? Why is it preferable than moving average? (10)
b) An analyst must decide between two different forecasting techniques for weekly sales of roller blades: a linear trend equation and the naïve approach. The linear trend equation is $F_t = 124 + 2t$, and it was developed using data from periods 1 through 10. Based on data for periods 11 through 20 as shown in the following table, which of these two methods is better? (17)

t	Units Sold	t	Units Sold
11	147	16	152
12	148	17	155
13	151	18	157
14	145	19	160
15	155	20	165

- c) Write down the three forecasting techniques and briefly explain them. (08)

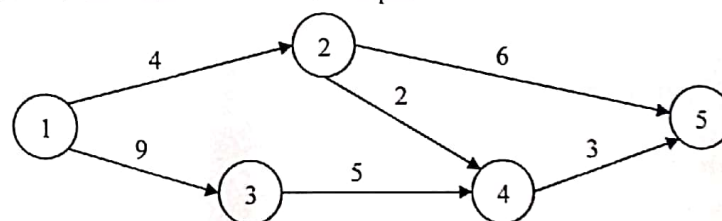
6. a) Briefly explain about the A-B-C approach to inventory control. (10)
 b) The Dine Corporation is both producer and a user of brass couplings. The firm operates 220 days a year and uses the couplings at a steady rate of 50 per day. Couplings can be produced at a rate of 200 per day. Annual storage cost is \$2 per coupling and machine setup cost is \$70 per run. Annual demand is 11,000 units. (17)
 i) Determine economic run quantity.
 ii) Approximately how many runs per year will there be?
 iii) Compute the maximum inventory level.
 c) Differentiate between level capacity strategy and chase demand strategy of aggregate planning. (08)
7. a) The product structure tree for end item E is given below. The manager wants to know the material requirements for part R that will be needed to complete 120 units of E by the start of week 5. Lead time for E is one week, one week for M and I, and two weeks for R, P, N and V. There is a scheduled receipt of 60 units of M at the start of week 2 and 100 units of R at the start of week 1. Lot for lot ordering is used. (17)



- b) Processing times and due dates for six jobs waiting to be processed at a work center are given in the following table. Determine the sequence of the jobs, average flow time, average tardiness and average number of jobs for each of these rules: (18)
 (i) FCFS, (ii) SPT and (iii) CR.

Job	Processing Time (days)	Due Date (days)
A	2	7
B	8	16
C	4	4
D	10	17
E	5	15
F	12	18

8. a) Write down the differences between product and process layout. (05)
 b) Using computing algorithm, determine the slack times for the following AoA diagram. Identify the activities that are on the critical path. (15)



- c) Develop an assignment plan that will minimize processing costs, given the information shown and compute the total cost. (15)

JOB	MACHINE		
	A	B	C
1	12	08	11
2	13	10	08
3	14	09	14
4	10	07	12

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
IEM 4127

Industrial Management

TIME: 1.5 hours

FULL MARKS: 120

- N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Define industrial management. Describe the process of management and explain how it can be used to accomplish results in any organization. (15)
b) Discuss critically the basic components of scientific management as propounded by Taylor. (15)
How far it is relevant in Bangladeshi situation?
2. a) Define the bureaucracy theory school. "A successful organization should react as an organic and open system". To what extent, you agree with this statement. Explain. (11)
b) Why the role of top management should be defined in decentralization? If it is not well defined, then what will be the implications? Explain. (13)
c) Write short notes on group dynamics and positive motivation. (06)
3. a) Discuss about the areas where Laissez-Faire leadership can be applied. (12)
b) An old car was purchased for 3,00,000 TK. Its life was estimated as ten years and the scarp value as 80,000 TK. Using the reducing balance method, calculate (18)
i) The depreciation rate (%).
ii) The depreciation fund at the end of three years.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) What is forecasting? What is the purpose of establishing control limits for forecast errors? (10)
b) The sale of mobile phone of a store over the last 8 years is shown in the table below. Plot the data, and visually check to see if a linear trend line would be appropriate. Then determine the equation of the trend line, and predict sales for years 2022 and 2023. (20)

Year	Unit Sales
2014	348
2015	370
2016	385
2017	355
2018	368
2019	390
2020	410
2021	425

5. a) Define project management. What is a breakdown structure and how it is useful for project planning? (10)
b) A manager just received a new price list from a supplier. It will now cost \$1.00 a box for order quantities of 801 or more boxes, \$1.10 a box for 200 to 800 boxes, and \$1.20 a box for smaller quantities. Ordering cost is \$80 per order and carrying costs are \$0.10 per box a year. The firm uses 3,600 boxes a year. The manager has suggested a "round number" order size of 800 boxes. The manager's rationale is that with a U-shaped cost curve that is fairly flat at its minimum, the differences in total annual cost between 800 and 801 units would be small anyway. How would you reply to the manager's suggestion? What order size would you recommend? (20)

6. a) What reasons usually prompt an existing organization to seek a new location? How can management avoid the problem of considering a new location? (10)
- b) Use the assignment method to determine the best way to assign workers to jobs, given the following cost information. Compute the total cost for your assignment plan. (20)

		JOB		
		A	B	C
Worker	1	5	8	7
	2	6	7	9
	3	4	5	3

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
IEM 4127

Industrial Management

TIME: 3 hours

FULL MARKS: 210

- N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define industrial management. Draw a network of management functions and explain it. (13)
b) "It is necessary to have management knowledge and management skill prior to an industrial engineer". Explain it in the sense of industrial management. (10)
c) Assume that you are a production manager in a reputed organization. Explain your managerial roles that are in the organization. (12)
2. a) Define job evaluation. Describe the factor comparison method of job evaluation with suitable example. (12)
b) A job is rated in terms of wages of Tk. 800 per day. The standard time set for the job is 10 days, 8 hrs/day. Two workers have taken 75 and 85 hours respectively for the completion of the job. A bonus of 70% on the time taken will be given only to those who have completed the job in the standard time. Calculate the earnings of each individual by Bedaux plan. Here, 1B stands for 0.8 standard work minute and it includes working time as well as time for rest. (15)
c) Differentiate between wages and salaries. Explain Bedaux plan with example. (08)
3. a) Define the bureaucracy theory school. "A successful organization should react as an organic and open system". To what extent you are agreeing with this opinion. Explain. (12)
b) Define group dynamics. Explain Hawthorne experiment regarding human nature and attitude. (10)
c) Differentiate delegation and decentralization. How can delegation be made more effective? (13)
4. a) Discuss Herzberg's theory of motivation. How does it differ from Maslow's theory? What are its limitations? (12)
b) Define upper and lower breakeven point. Why breakeven analysis is called short-run analysis? (10)
c) A cooling water pumping station at the LCRA plan costs \$600,000 to construct, and it is projected to have a 25 years life with an estimated salvage value of 15% of the construction cost. However, the station will be book depreciated to zero over a recovery period of 30 years. Calculate the annual depreciation charge for years 4, 10 and 25 using (i) straight line depreciation (ii) DDB depreciation (iii) implied salvage value for DDB. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define forecast. What advantages as a forecasting tool does experimental smoothing have over moving averages? (09)
b) For the following table of data, plot the data on graph and verify if a linear trend relation exists between them. If exists, then develop a line trend equation for the given data. Then use the equation to predict the next two values of the series. (13)

Period	Demand
1	44
2	52
3	50
4	54
5	55
6	55
7	60
8	56
9	62

- c) Write down the elements of a good forecast and briefly describe the steps in the forecasting process. (13)

6. a) Deduce the EOQ model with assumptions. Also, make a comparison between EOQ and EPQ (12)
model.
- b) A manufacturing firm uses roughly 3,400 pounds of chemical dye a year. Currently, the firm (13)
purchases 300 pounds per order and pays \$3 per pound. The supplier has just announced that
orders of 1,000 pounds or more will be filled at a price of \$2 per pound. The manufacturing
firm incurs a cost of \$100 each time it submits an order and assigns an annual holding cost of
17 percent of the purchase price per pound.
- i) Determine the order size that will minimize the total cost.
- ii) If the supplier offered the discount at 1,500 pounds instead of 1,000 pounds, what
order size would minimize the total cost?
- c) What is lead time? Describe briefly A-B-C approach to inventory control. (10)
7. a) Define MRP. Write short notes on (i) Master schedule (ii) Bill of materials (iii) Inventory (12)
records.
- b) A firm that produces wood shutters and bookcases has received two orders for shutters: one (15)
for 100 shutters and one for 150 shutters. The 100-unit order is due for delivery at the start of
week 4 of the current schedule, and the 150-unit order is due for delivery at the start of 8 week.
Each shutter consists of two frames and four slatted wood sections. The wood sections are
made by the firm, and fabrication takes one week. The frames are ordered and lead time is two
weeks. Assembly of the shutters require one week. There is a scheduled receipt of 70 wood
sections in week 1 (i.e., at the beginning). Determine the size and timing of planned-order
releases necessary to meet delivery requirements under lot-for-lot ordering condition.
- c) Write down the differences between product and process layout. (08)
8. a) Define CAPP. Briefly describe the roles of CAPP in manufacturing. (10)
- b) The following table contains information on the cost to run three jobs on four available (15)
machines. Determine an assignment plan that will minimize cost.

		Machine			
		A	B	C	D
Job	1	12	16	14	10
	2	9	8	13	7
	3	15	12	9	11

- c) A group of six jobs is to be processed through a two-machine flow shop. The first operation (10)
involves cleaning and the second involves painting. Applying Johnson's rule, determine a
sequence that will minimize the total completion time for this group of jobs. Processing times
are shown in the following table.

Processing time (hours)		
Job	Workcenter1	Workcenter2
A	5	5
B	4	3
C	8	9
D	2	7
E	6	8
F	12	15

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ii) Figures in the right margin indicate full marks.

iii) Standard Normal Distribution Table.

iv) Assume reasonable data if missing any.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define management. Why management is considered as a dynamic process? Explain. (13)
b) Describe the necessary skills that have to be possessed by a manager in brief. (12)
c) Explain how management and administration are similar and different. (10)
2. a) What is scientific management theory? Explain the Fayol's 14 principles of management. (15)
b) Explain contingency approach theory. What are the contribution and limitations of contingency approach? (15)
c) Write down three rules of manager. (05)
3. a) Define motivation. Discuss the two-factors theory of motivation. (15)
b) Explain the Murray's theory of human personality. (10)
c) Define the term "Recruitment". What are the differences between recruitment and selection? (10)
4. a) Define layout. State the layout format you would recommend and why, for: (i) A chemistry lab, (ii) A university cafeteria and (iii) The processing of mail at the post office. (10)
b) Is the facilities-location problem for services different from that for manufacturing and in what respects? Explain with example. (12)
c) Alum Co Ltd. Plans to set up a new aluminum-products plant and has selected three sites with following characteristics (for projected 75% capacity utilization equal to 300,000 tons/year) (13)

Site	Annual fixed costs (millions)	Variable costs/ton
A	8.2	\$452
B	10	290
C	7.5	530

Relevant Subjective factors	Rating R_{ij}			Relative importance index W_j
	A	B	C	
Housing	.25	.50	.25	.50
Community attitudes	.25	.25	.50	.25
Services	.50	.25	.25	.25

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define forecast. What are the main advantages that quantitative techniques for forecasting have over qualitative techniques? What limitations do quantitative techniques have? (12)
b) An electrical contractor's records during the last five weeks indicated the number of job requests. (13)

Week:	1	2	3	4	5
Requests:	20	22	18	21	22

Predict the number of requests for week 6 using each of these methods: (i) A linear trend equation, (ii) Naive, (iii) Exponential Smoothing with $\alpha = 0.30$. Use 20 for week 2 forecast.

- c) Define breakeven point graphically. Why break even study is important? Also explain, why breakeven quantity goes down when variable cost is reduced. (10)
6. a) "Inventory is called necessary evil" – explain. Describe briefly the A-B-C approach to inventory control. (12)
b) Deduce the EOQ model with assumptions. Also make a comparison between EOQ and EPQ model. (11)
c) A small manufacturing firm uses roughly 3400 pounds of chemical dye a year. Currently the firm purchases 300 pounds per order and pays \$3 per pound. The supplier has just announced that orders of 1000 pounds or more will be filled at a price of \$2 per pound. The manufacturing firm incurs a cost of \$100 each time it submits an order and assigns an annual holding cost of 17 percent of the purchase price per pound. (12)

(i) Determine the order size that will minimize the total cost.

(ii) If the supplier offered the discount at 1500 pounds instead of 1000 pounds, What order size would minimize total cost?

7. a) What is dummy activity? Why it is used? A project consists of a series of tasks labeled (10)
A, B, ..., H, I with the following relationships ($W < X$, Y means X and Y cannot start until W is completed; $X, Y < W$ means W cannot start until both X and Y are completed). With this notation construct the network diagram having the following constraints: $A < D$, E ; $B, D < F$; $C, E < G$; $B < H$; $F, G < I$.
- b) Why is there a need for aggregate planning? Briefly describe demand and capacity options for (10)
aggregate planning. Also discuss the basic strategies for meeting uneven demand.
- c) A project manager has compiled a list of major activities that will be required to install a (15)
computer information system in her firm. The list includes estimated completion times for activities and precedence relationships. If the project is finished within 26 weeks of it's start, the project manager will receive a bonus of \$1000, and if the project is finished within 27 weeks of it's start, the bonus will be \$500. Find the probability of each bonus.

Activity	Immediate Predecessor	Estimated times (Weeks)
A	—	2-4-6
D	A	6-8-10
E	D	7-9-12
H	E	2-3-5
F	A	3-4-8
G	F	5-7-9
B	—	2-2-3
I	B	2-3-6
J	I	3-4-5
K	J	4-5-8
C	—	5-8-12
M	C	1-1-1
N	M	6-7-11
O	N	8-9-13
END	H, G, K, O	

8. a) Write short notes on the following: (15)
(i) Market Value, (ii) Salvage Value, (iii) Preventive maintenance, (iv) Types of MIS, (v) Generative process planning.
- b) The following table contains information on the cost to run three jobs on four available (10)
machines. Determine an assignment plan that will minimize costs.

		Machine			
		A	B	C	D
Job	1	12	16	14	10
	2	9	8	13	7
	3	15	12	9	11

- c) Using the information given in the following table, determine an order processing sequence (10)
using s/o priority rule.

Order	Processing time remaining (days)	Due Date (days)	Number of operations remaining
A	20	30	2
B	11	18	5
C	10	6	2
D	16	23	4